CAPSTONE PROJECT 2

Final Report

The Role of Generative AI in Enhancing Student's Self-Regulated Learning and Academic Outcomes

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Abstract

The rapid adoption of generative AI tools such as ChatGPT and Gemini has transformed the educational landscape by providing students with personalised learning support, instant feedback and interactive assistance. However, limited empirical evidence exists regarding how the affordances of generative AI influence student's SRL and academic performance. Therefore, this study aimed to examine the relationships between generative AI affordances and SRL, investigate the relationship between SRL and perceived academic performance and validate a conceptual model examining these relationships using PLS-SEM. A quantitative cross-sectional research design was employed, and the data were collected through an online questionnaire from 222 students who actively use generative AI in their learning activities. The proposed conceptual model was developed based on Affordance Theory and Zimmerman's Self-Regulated Learning framework. The findings found that Planning Affordance positively influenced Task Analysis and Self-Motivation, while Feedback and Interaction Affordance positively influenced Self-Control, Self-Observation, Self-Judgement and Self-Reaction. Furthermore, Self-Motivation and Self-Observation were the only SRL dimensions that demonstrated significant positive relationships with perceived academic performance. The measurement model achieved satisfactory reliability and validity, while the structural model demonstrated acceptable explanatory and predictive capability. This confirms the validity of the proposed conceptual model. Overall, the findings indicate that generative AI contributes to improved academic performance primarily by strengthening student's SRL behaviours rather than directly improving learning outcomes.

Keywords — Self-regulated learning; Generative AI; affordance theory; Academic Performance; PLS-SEM

1.0 Introduction

1.1 Background Study

The education sector has undergone into a significant shift in recent years as the rapid growth of digital technologies. While the traditional method relied on instructor-led dissemination of knowledge, the modern educational method is more in emphasizing student-centred learning, where they must take an active role in their own academic development. In this case, the integration of Artificial Intelligence (AI) into the educational sector able to help student on their learning. The rapid growth of generative AI such as ChatGPT, Gemini and Copilot have provided a critical point in educational sector. The generative AI has introduced new opportunities for students to have personalized learning support. These models able to generate great explanations, summarize the complex concepts and provide instant feedback on student work like a tutor.

At the same time, self-regulated learning (SRL) is a critical skill that enables students to take control of their learning. SRL is defined as a cyclical process where the learner will set their goals, monitor their learning and regulate their motivation. Besides that, SRL involve some important phases such as planning and goal setting, monitoring their progress during the task and adjust their strategies based on the evaluation of the result (Panadero, 2017; Pintrich, 1995; Zimmerman, 2002). Thus, student who master un SRL are generally better to navigate the complex academic tasks and achieve better performance.

The relationship between generative AI affordances, SRL and academic performance has become an important area of research as generative AI is increasingly adopted by students in educational settings. Generative AI can support different phases of SRL by providing immediate feedback to facilitate student's monitoring of their learning progress and generate

study plans to assist with planning and goal setting. However, the effectiveness of these affordances depends on how students utilize the technology during their learning process. Students with stronger SRL skills are more likely to use generative AI effectively to support their learning. However, those with weaker SRL skills may not fully benefit from its capabilities. Thus, it is important to examine the relationships among generative AI affordances, SRL and academic performance to better understand how these factors interact and contribute to student's learning outcomes.

1.2 Problem Statement

As the rapid growth of AI in recent years, it has begun to significantly provide both opportunities and challenges in student education (Boubker, 2024). Among the most power tools emerging from this advancement are generative AI such as ChatGPT and Gemini, which are increasingly integrated into learning environments where they provide explanations, generate content, and offer instant feedback to students. Thus, these AI systems provide a strong potential to support student's SRL by creating an environment that allows them to plan, monitor and evaluate their learning progress to achieve academic goal.

Despite the rising adoption of generative AI in the education sector, there remains limited empirical evidence on how student's use these AI platforms to enhances their SRL behaviours to increase their academic performance. While previous studies from Silitubun (2023) have shown that AI tools can help identify student's strengths and weaknesses by tracking progress patterns during their SRL. While the other studies from Ge et al. (2025) had introduced a LLM-based system called "SRLAgent". They found that it significantly improved student's SRL skills by engagement through gamification and adaptive feedback.

Therefore, this study aims to address this research gap by examining the relationship between generative AI affordances and SRL. Other than that, the study also examine the relationship between SRL and student's academic performance, as well as the relationship between generative AI affordances and academic performance. Furthermore, the study develops and validates a conceptual model to examine the relationships among generative AI affordances, SRL and academic performance. By employing hypothesis testing and structural equation modelling techniques, the findings will provide empirical evidence on how generative AI affordances are associated with student's self-regulated learning behaviours and academic performance. The findings are expected to contribute to the effective integration of generative AI into educational practices and support students in achieving better learning outcomes.

1.3 Objectives

The objectives of this project are:

1. To assess the relationship between generative AI feedback usage and student self-regulated learning (SRL)
2. To examine the relationship between student's (SRL and academic performance.
3. To validate the proposed conceptual model examining the relationships among generative AI affordances, self-regulated learning, and academic performance using PLS-SEM.

1.4 Scope and Limitation

This study focuses on finding the relationship between the use of generative AI, SRL behaviours, and academic performance among students. The research will focus on student group that use generative AI in their SRL process. The scope of this study is to evaluate how generative AI influence student's SRL and academic outcomes.

However, the study has some limitations. Firstly, the data collected from participants may include bias as each participant have their own perception and understanding on SRL and AI

usage. Besides that, the finding will not be generalizable to all educational contexts as different participants have different accessibility to generative AI and SRL behaviour.

1.5 Research Significance

This study holds a significant value in enhancing student academic performance by linking SRL and generative AI. While AI tools are increasingly used in educational settings, their effectiveness in supporting student's SRL depend on how they use them to improve their processes. Therefore, there is a need to understand the interaction between SRL, generative AI usage and academic outcomes.

This study aims to bridge the gap between SRL and academic performance. The study seeks to provide theoretical insights into how generative AI can support student to plan, monitor and reflect on their progress. From a practical perspective, the finding of this study is expected to provide guidance to all on integrating generative AI into learning environment to enhance student's SRL.

2.0 Literature Review

2.1 Overview of SRL

SRL is the process where people plan, monitor and regulate their motivation and behaviour to create their own personal learning goals and achieve it. According to Zimmerman (2002), SRL is a cyclical process that transform mental abilities into academic skills. People who aware their strengths and limitations with a personal goal can improve their SRL development by enhancing their self-satisfaction and motivation. By doing this, they will have continuous self-awareness that helps them to refine their learning goals more effectively in the future. This will lead to academic growth and self-improvement. Besides that, Zimmerman (2002) further explains about the structure of the self-regulation process, which the people will go through three cyclical phases. The first phase is the forethought phase where involve task analysis and self-motivation. Besides that, the performance phase is the second phase where they apply self-control and self-observation to manage their attention, strategies and learning environment while doing a task. Finally, the self-reflection phase focuses on self-evaluation and self-reaction. The learner will assess their performance by identifying strengths and weaknesses. This provides the opportunity for them to improve their SRL in the future.

The other study from Pintrich (1995) defines SRL as a way of students approach their academic tasks through their experience and self-reflection. He highlighted that the characteristics of SRL have three aspects. First, the learner should be able to control their own behaviour so they can adjust their characteristics to fit a specific goal. Besides that, the learner should have a goal that provides them the standard to monitor their own performance and make suitable adjustments to achieve the goal. Lastly, they should have the ability to control their action and not respond to someone responses. In addition, the study from Panadero (2017) have compared six major SRL models from Zimmerman, Pintrich, Boekaert, Winne and Hadwin,

and Efklides. They have some similarities across all models where the learner must apply suitable learning strategies including goal setting, planning, monitoring and reflection to effectively manage their SRL. This shows the importance of learners regulating their own behaviour and motivation to achieve their learning goals.

In conclusion, SRL is a comprehensive process that requires learners to actively plan, monitor and regulate their cognition, behaviour and motivation to achieve their learning goals. It is a cyclical and strategic process where the learners need to set a goal and apply suitable learning strategies based on their strengths and limitations. Throughout the SRL progress, they need to keep monitor their own progress and reflect on their performance. This ongoing cycle will help learners to develop better control over their learning, support their overall academic development and enable continuous improvement in the future.

2.2 AI in SRL Development

SRL is a critical competency where students must take their responsibility in goal setting, planning, monitoring progress and reflecting on outcomes with minimal external supervision (Ge et al., 2025). Recent advances in generative AI like ChatGPT and Gemini have shown a strong potential to enhance SRL through personalized and adaptive support. Some studies highlight that AI will help learner identify their weaknesses, adjust their learning strategies and motivate them by providing a real-time explanations and reflective prompts (Ng et al., 2024). Unlike the traditional rule-based chatbots that rely on predefined scripts and keyword matching, LLM leverage large datasets to handle complex conversations. This provides the capability to allow AI model to become a “virtual teacher” by offering flexible and open-ended responses that encourage metacognitive reflection rather than providing fixed answer. For example, Ng et al. (2024) demonstrated that rule-based bots (Nemobot) are effective for rote practice, but

generative AI (SRLbot) is significantly effective in promoting behavioural engagement and motivation due to its ability to personalize recommendations based on specific learning scenarios. From their interactive dialogue, these models work as an intelligent tutor that can scaffold their cognitive and metacognitive processes in real time that allow them to improve their learning method.

The other research on the SRLAgent system illustrate how AI can structurally support Zimmerman's three phase SRL model. By utilizing specialized agents like Planning Agent, Subtask Tutor Agent System and Reflection Agent, the system able to provide immediate context-specific feedback that strengthens metacognitive awareness and self-monitoring behaviours (Ge et al., 2025). This approach will provide solutions that help students manage their task and allocate the time effectively.

Similarly, Shu and Xu (2022) found that AI-supported learning helps to train student's awareness in autonomous learning, which help them improves their self-reflection and allows them to adjust their learning strategies based on their individual weaknesses. Their research indicates that AI tools can foster student interest and curiosity, which essential for their sustaining motivation in self-directed environments. Furthermore, the frequency of interaction with AI agents is a significant predictor of SRL development. These mean students who actively engage with generative AI tend to develop a better self-regulation strategy compared to other.

Overall, personalized and adaptive feedback serves as the key mechanism through which AI enhance SRL, thereby establishing the foundation for improved academic performance discussed in the subsequent section.

2.3 AI and Academic Performance

Recent studies have found that generative AI like ChatGPT and Gemini are altering student's approaches to learn with the increase on evidence of their impact on academic performance. Several studies shows that LLM-assisted environments can significantly improve student achievement and engagement by offering personalized explanations and interactive support. For example, Ahmed et al. (2025) created an LLM-based system that provided personalized suggestion that help to enhance student's learning performance and well-being. Their results showed that the LLMs personalised feedback give a strong potential in helping students enhance their academic outcomes. However, they had highlighted that there is a need of an empirical validation and statistical testing to confirm these effects toward the academic success. Similarly, Kumar et al. (2024) examined how students use AI by doing two studies that compared four pedagogically based guidance strategies. While structured guidance reduced random queries and instances of copy-pasting assignment questions, the improvement in learner performance was not significant. These findings show the difficulty of measuring AI impact and show how students engage with AI may be as important as simply having access to them. In addition, Shehri et al. (2023) found a positive relationship between the use of ChatGPT and the student's learning quality, creativity, and overall academic performance improvement. The study also identified prompt engineering as a mediating factor, where the effectiveness of AI is not only influenced by access to the technology but also by how learner interact with it. Overall, these studies had found that AI have the potential to enhance student's academic outcomes by providing personalized and interactive support. However, their effectiveness depends on how learners use well with the technology.

2.4 Challenges of AI Integration

2.4.1 Over-reliance on AI

Although AI can be a good tool to enhance student SRL. It also brings concern as the student may too rely on the result generate by them. Based on Bo et al. (2025) describe how learners tend to over-rely on AI-generated responses, leading to superficial learning as they do not understand and the metacognitive monitoring is reduced. Their findings show that constant access to ready-made explanations encourages passive consumption rather than active inquiry Thus, this over reliance are limiting the opportunities for reflection and self-evaluation of a student. Another study from Zhai et al. (2024) found that AI dialogue systems had introduce risk to student's critical thinking and diminishing their creativity. Students may bypass the cognitive work that needed to analyse, evaluate and generate original ideas as they just accept the AI output without understanding. In addition, Kim et al. (2025) found that over-reliance on AI provide issue and risk to the user who trust the incorrect responses. They find the AI responses increase user's reliance when it provides explanation for both correct and incorrect answers. This causes the user over-reliance the result without proving it from other source. In short, while AI provide significant benefit in supporting student's SRL, they still provide risk when the student depends heavily on the context generated by AI that may weaken their critical thinking, creativity and metacognitive awareness.

2.4.2 Accuracy and Bias in AI Responses

The other challenges for AI where they may produce misleading and inaccurate information. These inaccuracies can negatively affect student's understanding and decision-making during SRL. Based on the research from Spatharioti et al. (2025), their finding concludes that AI provide accuracy of 95% for LLM-based search for the task for the easier task. However, the AI accuracy for challenging task dropped significantly to 47%. For SRL

progress, students sometimes focus on complex questions that will increases the risk of receiving incorrect or inaccurate explanations and answers from AI. Thus, this can negatively impact their learning outcomes and overall effectiveness of their SRL process. The other study from (Zhou et al., 2024) found that advanced models like GPT-4 try to generate inaccurate answer with detail explanation instead of refuse to respond when they are uncertain. This overconfidence creates a false sense of accuracy, that cause users to trust wrong outputs. These errors can affect learners as they may learn misinformation without knowing. The challenges of misleading and inaccurate information are also closely linked to algorithmic bias that affect the LLM output. These biases do not come from a single source, but rather throughout the entire LLM life cycle (Lee et al., 2024). The study by Warr et al. (2024) proved that LLMs show a damaging form of implicit bias in objective tasks. The models showed significant score variations when a student's race was indirectly inferred using socioeconomic and school-type variables ultimately leading to scores that reflected existing patterns of systemic inequity. Moreover, Weissburg et al. (2024) describe LLMs function as "biased teachers" when it tasked with personalized education. The study had found that significant biases across different demographic groups, including income, disability status, race, and gender. In conclusion, the accuracy and bias challenges of AI provide significant risks to students engaging in SRL. Although the AI can generate useful and contextually relevant responses, but their tendency to produce incorrect or biased information may lead to misconceptions and affect their critical thinking development.

3.0 Methodology

3.1 Affordance Theory

Affordance theory describes the potential actions that an environment or object offers to an organism, shaped by the individual's capabilities and the context of interaction (Pan, 2026). In the current digital technologies world, affordances are not merely inherent features of a system but emerge through the interaction between users and technological artefacts. This means that the same technology may enable different actions depending on how the user used it. In educational settings, generative AI provides various affordances such as supporting task planning, enabling interactive feedback, and facilitating reflective learning. Meanwhile, Zhuang et al. (2025) describe AI affordances as the actionable capabilities that AI provides to its users. These affordances are realized through specific digital tools which facilitate the execution of distinct pedagogical tasks. These affordances help to engage student in their learning processes by helping them to organise study tasks, monitor their understanding and evaluate their learning progress. Furthermore, AI affordances support personalised and adaptive learning experiences by allowing students to receive guidance and feedback based on their needs to enhance their SRL behaviours.

3.2 Conceptual Model

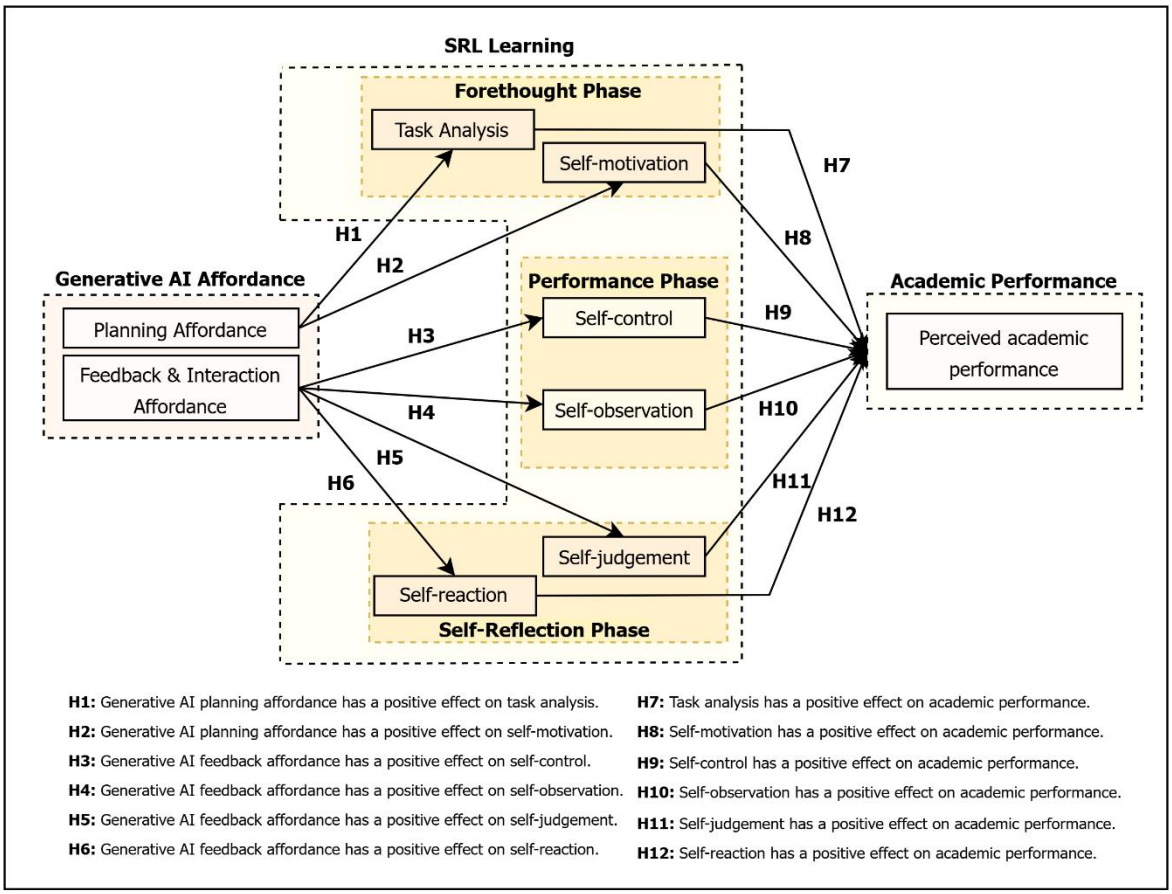
This study proposes a conceptual model to examine how generative AI support student's SRL process and influences academic performance. The model have applied Zimmerman SRL framework by introducing forethought, performance and self-reflection phases. Within this framework, generative AI is an external learning support that may enhance their planning, monitoring and reflective behaviours during their SRL.

The model also investigates two categories of generative AI affordance, which is study

planning affordance and feedback and interaction affordance, and their effect on key SRL attributes. In addition, the model also examines how the SRL attributes influence academic performance. This study position SRL as the mediating role to explain how generative AI affordance contribute to academic outcomes. Figure 1 shows the proposed conceptual model of this study.

Figure 1

Conceptual Model



3.2.1 Generative AI Affordance

3.2.1.1 Planning Affordance

Planning affordance is the capability of generative AI to support students in organising learning tasks, setting goals and selecting suitable learning strategies before engaging in

academic activities. According to Zimmerman (2002), effective learning begins during the forethought phase, where planning and goal setting play crucial roles in preparing students for successful task execution. Generative AI tools can support this process by assisting students in breaking down learning tasks, generating study outlines, clarifying task requirements, and suggesting appropriate learning strategies based on students' needs. Therefore, planning affordance is expected to positively influence task analysis and self-motivation within the forethought phase of SRL.

The following are the proposed hypotheses related to planning affordance:

H1: Generative AI planning affordance has a positive effect on task analysis.

H2: Generative AI planning affordance has a positive effect on self-motivation.

3.2.1.2 Feedback & Interaction Affordance

Feedback and interaction affordance is the capability of generative AI to provide explanations, clarification, feedback and interactive support during and after learning activities. Within the SRL framework, feedback and monitoring support are important elements in the performance and self-reflection phases (Pintrich, 1995; Zimmerman, 2002). Generative AI allows students to ask questions, receive immediate explanations, identify mistakes and obtain suggestions for improvement. This interactive support enables students to monitor their understanding, regulate their learning behaviours, and reflect on their performance more effectively. Therefore, feedback and interaction affordance is expected to positively influence self-control, self-observation, self-judgement, and self-reaction.

The following are the proposed hypotheses related to feedback and interaction affordance:

H3: Generative AI feedback and interaction affordance has a positive effect on self-control.

H4: Generative AI feedback and interaction affordance has a positive effect on self-observation.

H5: Generative AI feedback and interaction affordance has a positive effect on self-judgement.

H6: Generative AI feedback and interaction affordance has a positive effect on self-reaction.

3.2.2 Forethought Phase

The forethought phase start before the actual learning task begins. It refers to preparatory processes that shape the learner's intention and foundation for their effective learning. This phase consists of two main categories, which is task analysis and self-motivation. In task analysis, it includes goal setting and strategic planning where thew student had set a specific goal for themselves can increase their academic success (Zimmerman, 2002). Self-motivation refers to the self-beliefs that control their learning behaviour such as self-efficacy, interest, and the perceived value of the task. Thus, these elements shape how effectively students initiate and approach their learning activities.

The following are the proposed hypotheses related to forethought phase:

H7: Task analysis has a positive effect on academic performance.

H8: Self-motivation has a positive effect on academic performance.

3.2.2.1 Task Analysis

Task analysis is the learner's ability to break down a learning task into manageable steps, set goals and develop appropriate strategies for accomplishing those goals. In this study, task analysis will examine how the students define their learning objectives, plan their study approach and prepare themselves before using with generative AI. Besides that, effective task analysis will help learners to allocate time, select strategies and position themselves for successful learning outcomes.

3.2.2.2 Self-Motivation

Self-motivation includes the beliefs and attitudes that affect a learner's interest to engage in a task. It includes self-efficacy, intrinsic interest and the perceived relevance of the task based on their interest. This study evaluates self-motivation by measuring how confident, interested and driven students feel before beginning their learning activities. Higher self-motivation will increases the likelihood that students will use generative AI purposefully to support their goals to ensure accuracy and progress.

3.2.3 Performance Phase

The performance phase will happen during the learning task is carried out. It involves two categories, self-control and self-observation. Self-control includes the focusing attention and deploying specific strategies that focusing attention on the task to optimize their learning such as self-instruction, task strategies, imagery and attention focusing (Zimmerman, 2002). Besides that, self-observation need the learner systematically tracking their own performance to detect the mistake they make by doing self-recording and self-experimentation. In this study, the performance phase will assess how students manage their attention, avoid distraction and track their own progress during they are studying with generative AI support.

The following are the proposed hypotheses related to performance phase:

H9: Self-control has a positive effect on academic performance.

H10: Self-observation has a positive effect on academic performance.

3.2.3.1 Self-Control

Self-control refers how students keep themselves to stay focus during their learning. It

involves using strategies in avoid distraction during their learning and remind themselves to stay on their planning track. In this study, this will examine how the students will regulate their learning behaviour and maintain attention while interacting with generative AI to support their SRL.

3.2.3.2 Self-Observation

Self-observation is the process of monitoring their performance during their SRL. They need to keep tracking of their progress by identifying error and evaluate their strategy effectiveness to make real-time adjustments. In this study, this captures how students check their understanding and verify the AI-generated responses to their learning goal.

3.2.4 Self-reflection Phase

The self-reflection phase will happen after their learning task is completed. This phase will involve self-judgement and self-reaction. Self-judgement involves self-evaluation and casual attribution as the learner need to compare their performance and determines the reason of getting the results. Besides that, self-reaction involves making adaptive inferences and behavioural adjustments for future tasks**.**

The following are the proposed hypotheses related to self-reflection phase:

H11: Self-judgement has a positive effect on academic performance.

H12: Self-reaction has a positive effect on academic performance.

3.2.4.1 Self-Judgement

Self-judgement involves assessing the quality of one's performance through self-evaluation and casual attribution. The learner will compare their outcomes against expected

targets and identify factors that contributed to their results. In this study, this will examine how student evaluate their learning outcomes with the support of the generative AI feedback and how they determine the causes of their performance.

3.2.4.2 Self-Reaction

Self-reaction refers to the learner's emotion and response after they evaluate their performance. This includes how they modify their learning strategies and planning strategies for their future SRL. In this study, this will measure how students will adjust their future study approaches based on their reflections and their interactions with generative AI.

3.2.5 Academic Performance

Academic performance refers to student's overall learning outcomes and often used as an indicator of learning effectiveness. Previous study has consistent shown that student who integrate with AI into learning can increase their academic performance and motivation (Silitubun, 2023). This happened as generative AI can provide instant feedback and personalised guidance that support student's understanding. Besides that, it often represents as an outcome of learner's ability to effectively plan, monitor and reflect on their learning processes (Zimmerman, 2002). Thus, this study treats academic performance as outcome variable to examine which factors influence student's academic performance.

3.3 Research Design

This study will use quantitative, cross-sectional correlational research design to examine the structural relationships between effective generative AI, SRL and academic performance. This research design is appropriate because it enables the examination of the relationships

among multiple latent constructs simultaneously using SEM. Furthermore, a deductive research approach will be employed to test the proposed conceptual framework grounded in Affordance Theory and Zimmerman's SRL framework. The design enables the study to empirically investigate the relationships between generative AI affordances, student's SRL process and academic performance.

3.4 Population and Sampling

The target population of this study involve high school, college and university students as these groups frequently used a lot of generative AI tools such as ChatGPT and Gemini to support their learning task. A snowball sampling method is used as it provides feasibility and accessibility considerations when finding respondents. The study aims to collect 200 student's responses. This sample size decision is supported by Westland (2015) who report that SEM with three indicators will require a minimum of 200 sample size to achieve a stable parameter estimates and proper model convergence. Therefore, a sample of 200 will give sufficient statistical power for the proposed analysis.

3.5 Questionnaire Design

The data will be collected through an online questionnaire using Google Forms. The instrument will consist of four sections, which is demographic question section SRL section, generative AI usage strategy section and academic performance section. Besides that, ethical considerations are addressed through a digital consent form presented at the beginning of the questionnaire to inform participants where their participation is voluntary and the responses will remain anonymous to be used for research purposes only. To ensure the data security, all responses will only be stored in my own device with password to ensure only me can access.

This study will use the 5-point Likert Scale for most of the section in this questionnaire. According to Koo and Yang (2025), Likert scale is a rating scale that measure unobservable constructs like opinion and perception through a questionnaire. It allows the responder to rate their level of agreement with a 5-point scale ranging from “strongly disagree” to “strongly agree”. Furthermore, this measurement method is effective to minimize the respondent fatigue compared to the larger scales’ method such as MSLQ while maintain a high reliability and validity. With 5-point Likert Scale, we able to use SEM model in this research.

Table 1

Measurement Items using 5-point Likert Scale

5-point Likert scale	**Item No.**
1 of 5	Strongly Disagree
2 of 5	Disagree
3 of 5	Neutral
4 of 5	Agree
5 of 5	Strongly Agree

3.5.1 Demographic Questions

The section A will be collecting the background information to establish the profile of the respondents is relevant to the research

Table 2

Demographic Questions

Question	Reference
What is your gender? - Male - Female	(Qian, 2025)
What is your age? - 16 – 18 - 19 – 25 - 26 – 30 - 31 and above	(Qian, 2025)
What is your education level? - Secondary School (High School) - Pre-U (Foundation, MUFY) - Diploma - Undergraduate (bachelor's degree) - Postgraduate	(Qian, 2025)

3.5.2 Measurement Items

Table 3

Section A: Affordance Theory

Item No.	Question	Reference
PA_1	I use generative AI to help me plan what I need to study before starting a task.	(Daher & Abdela, 2025)
PA_2	Generative AI help me break down learning tasks into manageable steps.	(Daher & Abdela, 2025)
PA_3	I use generative AI to clarify the requirements of a task before beginning.	(Daher & Abdela, 2025)
PA_4	Generative AI help me decide suitable learning strategies for my study goals.	(Daher & Abdela, 2025)
PA_5	I use generative AI to organise my study schedule or learning priorities.	(Daher & Abdela, 2025)
FIA_1	I use AI explanations to help me understand that I do not know.	(Daher & Abdela, 2025)
FIA_2	AI tools help me identify mistakes in my work.	(Daher & Abdela, 2025)
FIA_3	I use AI tools to get suggestions on the way to improve my answers	(Daher & Abdela, 2025)
FIA_4	Generative AI provides useful feedback that helps me improve my learning.	(Pintrich et al., 1991; Daher & Abdela, 2025)
FIA_5	Generative AI helps me reflect on the effectiveness of my	(Ng et al., 2024)

learning strategies.

Table 4

Section B: Self-regulated learning

	Item No.	Question	Reference
Forethought Phase	TA_1	I think about what I really need to learn before I begin my self-regulated learning with the help of generative AI.	(Jansen et al., 2017)
	TA_2	I set short-term goals as well as long-term goals for my own self-regulated learning with the help of generative AI.	(Jansen et al., 2017)
	TA_3	I set specific goals before I begin a task with the help of generative AI.	(Broadbent et al., 2023; Jansen et al., 2017)
	TA_4	I analyse in depth the task I have to complete so that it is clear to me what I have to do with the help of generative AI.	(Panadero et al., 2021)
	SM_1	I'm confident I can understand the most difficult learning material during my self-regulated learning with the help of generative AI.	(Daher & Abdela, 2025)
	SM_2	Even when study materials are uninteresting, I keep working until I finish with the help of generative AI.	(Pintrich & De Groot, 1990)
	SM_3	I persuade myself to work hard in SRL to improve my knowledge with the help of generative AI.	(Shen & Wang, 2025)

	SM_4	I try to connect the learning task to my personal interest with the help of generative AI.	(Shen & Wang, 2025)
Performance Phase	SC_1	I make sure I keep up with the weekly plan of my self-regulated learning with the help of generative AI.	(Jansen et al., 2017)
	SC_2	I try to avoid distractions when I am studying with the help of generative AI.	(Broadbent et al., 2023)
	SC_3	I use suitable learning strategies to improve my understanding during learning with the help of generative AI.	(Jansen et al., 2017)
	SC_4	I manage my study time carefully during my self-regulated learning activities with the help of generative AI.	(Jansen et al., 2017)
	SO_1	When I work on a task, I stop to check if I am progressing as planned with the help of generative AI.	(Panadero et al., 2021)
	SO_2	I ask myself questions about how well I am doing during my self-regulated learning with the help of generative AI.	(Jansen et al., 2017)
	SO_3	I am aware of what strategies I use when I study with the help of generative AI.	(Jansen et al., 2017)
	SO_4	While I perform a task, I check if the steps I am taking are appropriate with the help of generative AI.	(Panadero et al., 2021)
Self-Reflection	SJ_1	I think about what I have learned after I finish my SRL task with the help of generative AI.	(Jansen et al., 2017)
	SJ_2	I ask myself how well I accomplished my goals once I'm finished my SRL.	(Jansen et al., 2017)

SJ_3	I compare my learning outcomes with the goals I set before learning with the help of generative AI.	(Jansen et al., 2017)
SJ_4	I find myself analyzing the usefulness of my learning method while I study with the help of generative AI.	(Jansen et al., 2017)
SR_1	I adjust my learning strategies based on feedback with the help of generative AI.	(Pintrich et al., 1991; Broadbent et al., 2023)
SR_2	I use past mistakes to improve future learning with the help of generative AI.	(Pintrich et al., 1991; Broadbent et al., 2023)
SR_3	After the learning task completed, I asked myself on the way to improve in the next learning with the help of generative AI.	(Ebbes et al., 2024)
SR_4	I become motivated to learn other topics when my academic performance is improved with the help of generative AI.	(Shen & Wang, 2025)

Table 5

Section C: Perceived Academic Performance

Item No.	Question	Reference
AP_1	I am confident that I can understand what I study and gain better performance in the exam.	(Ferla et al., 2010)

AP_2	I am able to complete my academic tasks effectively after my studies.	(Ferla et al., 2010)
AP_3	I believe I can achieve good academic performance in my studies。	(Boubker, 2024)
AP_4	I am satisfied with my overall academic performance after my studies.	(Boubker, 2024)

3.6 Structural Equation Modelling (SEM)

SEM is a comprehensive multivariate analysis method that examine the complex relationship between constructs and indicators (Hair et al., 2021). Unlike the basic regression analysis, SEM will help in investigating the unique influence of one variable on another while simultaneously accounting for measurement errors and controlling for the influence of related variables (Zheng et al., 2021). The application of SEM basically involves two step approaches. Firstly, it needs to validate the measurement model and followed by the test of the structural model. Based on Vogelsmeier et al. (2025), the first step often uses confirmatory factor analysis (CFA) to test if the theoretical factor structure of survey instruments such as MSLQ can fits the collected data. This help to ensure the survey items accurate reflect underlying psychological constructs before causal relationship is tested. After the constructs are validated, the SEM is evaluated using specific Goodness-of-Fit (GoF) indices to determine if the hypothesized paths match the observed data. Based on Lim et al. (2020) and Zheng et al. (2021), a model generally considered by observe their Chi-square, comparative fit index (CFI) and a Root Mean Square Error of Approximation (RMSEA). For example, Lim et al. (2020) had used SEM to obtain the relationship between peer learning, SRL and academic performance where the SEM fit the GoF (chi-square = 1.697; CFI = 0.916; RMSEA = 0.045). Similarly, Zheng et al. (2021) used SEM to examine the relationships among self-efficacy, academic motivation, self-regulated learning

and academic achievement. Their findings demonstrated that SEM is an effective analytical approach for evaluating complex relationships among multiple latent constructs. Similarly, the current study adopts SEM to examine the relationships among generative AI affordances, SRL and student's academic performance. By using SEM, it enables the simultaneous assessment of both the measurement model and the structural relationships among the proposed constructs.

3.7 Statistical Analysis

The collected data will be analysed using Partial Least Squares Structural Equation Modelling (PLS-SEM). PLS-SEM is suitable for this study because it allows the assessment of complex relationships among latent constructs. Besides that, it is suitable for predictive research models with multiple constructs and indicators (Hair et al., 2021). Furthermore, PLS-SEM is highly suitable for analysing data collected from Likert scale questionnaires, which may not follow a normal distribution. In addition, it can estimate complex structural models that involve multiple independent and dependent variables simultaneously. This makes PLS-SEM an appropriate analytical technique for find out the relationships between generative AI usage, SRL and academic performance in this study.

3.8 PLS-SEM

3.8.1 Pilot Testing

A pilot testing is required before the main data collection to ensure the clarity, reliability and the validity of the questionnaire items. The pilot study will involve about 10 respondents who share similar characteristics with the target population to identify ambiguous wording and assess the reliability of the construct so it can be improved better.

The reliability of each construct is examined using Cronbach's alpha, where values above 0.70 indicate acceptable internal consistency (Hair et al., 2021). Based on the pilot test results, questionnaire items with low reliability or unclear wording may be revised or removed to improve the overall quality of the instrument before proceeding with the main data collection.

3.8.1.1 Assessment of Measurement Model

The measurement model assessment evaluates the reliability and validity of the constructs used in the study. Since all constructs in this study are reflective, the assessment follows standard PLS-SEM criteria.

Indicator Reliability

Indicator reliability is the first step that assessed by examining the outer loadings of each measurement item. The indicator loading above 0.7 are recommended as they able to explains more than 50 percent of the variance in the construct (Hair et al., 2021). In the other way, indicator loading between 0.40 and 0.70 is removed if they do not affect construct reliability and validity (Hair et al., 2021).

Internal Consistency Reliability

Internal Consistency Reliability is the second step that measure the same construct are associated with each other using composite reliability and Cronbach's alpha (Hair et al., 2021). The reliability values between 0.6 to 0.7 is acceptable in exploratory research.

Convergent Validity

The third step is convergent validity that used to evaluate a construct's convergent validity by using the Average Variance Extracted (AVE) (Hair et al., 2021). The minimum acceptable AVE is 0.5 which mean the construct explain 50% of indicator variance that make up the

construct.

Discriminant Validity

The fourth step is discriminant validity that used to measure which construct is empirical distinct from other constructs in the structural model (Hair et al., 2021). Heterotrait–Monotrait ratio (HTMT) is the method to measure discriminant validity. HTMT values below 0.85 indicate adequate discriminant validity between constructs, ensuring that each construct measures a distinct concept.

3.8.1.2 Assessment of Structural Model

The assessment of the structural model focuses on examining the relationships between latent constructs and evaluate the model predictive capability. This involve evaluating collinearity issues, significance and relevance of path relationships, explanatory power, effect sizes and predictive power.

Collinearity Issues

Collinearity issues will happen in regression model that may cause a significant regression coefficient to appear statistically insignificant (Kim, 2019). Thus, it will be examined for structural model regressions by calculate the variance inflation factor (VIF) values (Hair et al., 2021). VIF values above 5 is considered as potential collinearity issues. Therefore, VIF values below 5 are considered as acceptable for the structural model.

Assess the Significance and Relevance of the Structural Model Relationships

The significance of the hypothesis relationships is evaluated using path coefficients (β) (Hair et al., 2021). Path coefficients are the strength and direction of the relationship between

exogenous and endogenous construct. They typically range between -1 and +1 where values closer to negative 1 mean strong negative relationship. At the same time, value closer to positive 1 is strong positive relationship. A path coefficient is considered statistically significant at the 5% significance level when the confidence interval does not include zero. Significant path coefficients provide empirical support for the proposed hypotheses.

Explanatory Power and Effect Size

The explanatory power of the structural model is assessed using the coefficients of determination (R^2) of the endogenous construct (Hair et al., 2021). The R^2 value will indicate the proportion of variance in the endogenous construct explained by its predictor construct. The higher R^2 value mean a stronger explanatory power of a model. In addition, the effect size (f^2) uses to assess the impact of the change in R^2 when a specific predictor construct is removed from the model. Based on Hair et al. (2021), the higher f^2 mean more larger effect sizes.

Predictive Power

To assess the model's predictive power, PLSpredict is used to evaluate out-of-sample prediction performance using k-fold cross-validation (Hair et al., 2021). It assesses the model's ability to predict new or unseen observations. First, the model's predictive relevance is evaluated using **Q^2_predict**. When the values is greater than zero, this indicate that the PLS-SEM model has predictive relevance (Shmueli et al., 2019). Subsequently, predictive accuracy is evaluated using prediction error measures such as Root Mean Square Error (RMSE) and Mean Absolute Error (MAE). Lower RMSE (or MAE) values indicate higher predictive accuracy of the structural model. When the PLS-SEM model produces lower prediction errors than the LM for all indicators, the model demonstrates high predictive power (Shmueli et al., 2019).

Lower RMSE and MAE values mean higher predictive accuracy of the structural model.

3.8.2 Pilot Testing Measurement Model

Table 6

Pilot Testing Measurement Model

Construct	Item	Convergent Validity		Internal Consistency Reliability	
		Outer loadings	Average Variance Extracted (AVE)	Cronbach's Alpha	Composite Reliability
Planning Affordance	PA_1 ← PA	0.789	0.421	0.654	0.776
	PA_2 ← PA	0.661			
	PA_3 ← PA	0.538			
	PA_4 ← PA	0.425			
	PA_5 ← PA	0.757			
Feedback & Interaction Affordance	FIA_1 ← FIA	0.803	0.527	0.698	0.816
	FIA_2 ← FIA	0.845			
	FIA_3 ← FIA	0.904			
	FIA_4 ← FIA	0.679			
	FIA_5 ← FIA	0.006			
Forethought Phase	TA_1 ← TA	-0.291	0.306	0.316	0.355
	TA_2 ← TA	0.767			
	TA_3 ← TA	0.742			
	TA_4 ← TA	0.017			
	SM_1 ← SM	0.616	0.539	0.599	0.781
	SM_2 ← SM	0.963			

	SM_3 ← SM	0.920			
	SM_4 ← SM	0.068			
Performance Phase	**SC_1 ← SC**	0.774	0.425	0.500	0.683
	SC_2 ← SC	0.029			
	SC_3 ← SC	0.917			
	SC_4 ← SC	0.509			
	SO_1 ← SO	-0.015	0.259	0.603	0.213
	SO_2 ← SO	-0.193			
	SO_3 ← SO	0.113			
	SO_4 ← SO	0.992			
Self-Reflection Phase	**SJ_1 ← SJ**	0.870	0.439	0.576	0.721
	SJ_2 ← SJ	0.338			
	SJ_3 ← SJ	0.311			
	SJ_4 ← SJ	0.888			
	SR_1 ← SR	0.915	0.760	0.894	0.927
	SR_2 ← SR	0.849			
	SR_3 ← SR	0.928			
	SR_4 ← SR	0.788			
Perceived Academic Performance	**AP_1 ← AP**	0.871	0.714	0.867	0.910
	AP_2 ← AP	0.896			
	AP_3 ← AP	0.816			
	AP_4 ← AP	0.798			

The measurement model was assessed by evaluating their indicator reliability, internal consistency reliability and convergent validity. The assessment was conducted using SmartPLS 4. **Table 6** presents the measurement model assessment results.

Indicator Reliability

For **planning affordance** construct, PA_1 (0.789) and PA_5 (0.757) achieved the target outer loading values, PA_2 (0.661) was slightly below the recommended threshold. However, PA_3 (0.538) and PA_4 (0.425) showed a weak loading that require for further revision. For **feedback and interaction affordance** construct, FIA_1 (0.803), FIA_2 (0.845) and FIA_3 (0.904) had meet the recommended threshold, while FIA_4 (0.679) was slightly below 0.7. FIA_5 (0.006) had provided an extremely low outer loading, which may need revision later. For the **forethought phase**, TA_2 (0.767), TA_3 (0.742), SM_2 (0.963), SM_3 (0.920) achieved the required outer loading values. Meanwhile, TA_1 (-0.291). TA_4 (0.017), SM_1 (0.616) and SM_4 (0.068) did not meet the expected threshold. For the **performance phase**, SC_1 (0.774), SC_3 (0.917) and SO_4 (0.992) meet the recommended threshold. While SC_2 (0.029), SC_4 (0.509), SO_1 (-0.015), SO_2 (-0.193) and SO_3 (0.113) did not meet the recommended threshold. For the self-reflection phase, SJ_1 (0.870), SJ_4 (0.888), SR_1 (0.915), SR_2 (0.849), SR_3 (0.928) and SR_4 (0.788) meet the recommended threshold. Meanwhile SJ_2 (0.338) and SJ_3 (0.311) demonstrated a weak loading and need to be adjust later. Finally, the **perceived academic performance** construct demonstrated excellent indicator reliability. All four indicators exceeded the recommended threshold, with outer loading values ranging from 0.798 to 0.896.

Internal Consistency Reliability

Among all constructs, **perceived academic performance** achieved the highest internal consistency reliability, where the Cronbach's Alpha value is 0.867 and composite reliability is 0.910, which indicate this construct is excellent internal consistency. The **self-reaction** construct also showed a strong reliability where its Cronbach's Alpha is 0.894 and Composite Reliability of 0.927. The **feedback and interaction** affordance have a composite reliability of

0.816 and Cronbach's Alpha of 0.698, where the Cronbach's Alpha is slightly below the threshold but still acceptable. Similarly, **planning affordance** fulfil the composite reliability threshold (0,776) but have a low Cronbach's alpha (0.654). It also similar for **self-motivation** construct (Cronbach's Alpha of 0.599 and Composite Reliability of 0.781) and **self-judgement** (Cronbach's Alpha of 0.576 and Composite Reliability of 0.721), will meet the composite reliability threshold but not the Cronbach's Alpha. Furthermore**, task analysis** construct, **self-control** construct and **self-observation** construct demonstrated a poor internal consistency, which did not meet the threshold for both. By having these result, it provides the insight to find which item require adjustment before starting collect the real result from others.

Convergent Validity

The results showed that **Perceived Academic Performance** (AVE = 0.714), **Feedback and Interaction Affordance** (AVE = 0.527), **Self-Motivation** (AVE = 0.539) and **Self-Reaction** (AVE = 0.760) achieved the acceptable convergent validity. However, **Planning Affordance** (AVE = 0.421), **Task Analysis** (AVE = 0.306), **Self-Control** (AVE = 0.425), **Self-Observation** (AVE = 0.259) and **Self-Judgement** (AVE = 0.439) did not achieve the recommended threshold of 0.50.

3.8.3 Adjusted Measurement Items

Table 7

Section A: Affordance Theory (Adjusted)

Item No.	Question	Reference
PA_1	I use generative AI to help me plan what I need to study before starting a task.	(Daher & Abdela, 2025)

PA_2	Generative AI help me break down learning tasks into manageable steps.	(Daher & Abdela, 2025)
PA_3	I use generative AI to help me understand what is required to complete a learning task.	(Daher & Abdela, 2025)
PA_4	I use generative AI to suggest effective ways to study to achieve my learning goals.	(Daher & Abdela, 2025)
PA_5	I use generative AI to organise my study schedule or learning priorities.	(Daher & Abdela, 2025)
FIA_1	I use AI explanations to help me understand that I do not know.	(Daher & Abdela, 2025)
FIA_2	AI tools help me identify mistakes in my work.	(Daher & Abdela, 2025)
FIA_3	I use AI tools to get suggestions on the way to improve my answers	(Daher & Abdela, 2025)
FIA_4	Generative AI provides useful feedback that helps me improve my learning.	(Pintrich et al., 1991; Daher & Abdela, 2025)
FIA_5	Generative AI helps me understand whether my current study methods are effective.	(Ng et al., 2024)

Table 8

Section B: Self-regulated learning (Adjusted)

	Item No.	Question	Reference
Fo	TA_1	Before I start studying, I use generative AI to identify	(Jansen et al.,

		the topics I need to learn.	2017)
	TA_2	I use generative AI to help me set short-term and long-term learning goals.	(Jansen et al., 2017)
	TA_3	I set specific goals before I begin a task with the help of generative AI.	(Broadbent et al., 2023; Jansen et al., 2017)
	TA_4	I use generative AI to break down learning tasks into clear steps before I begin.	(Panadero et al., 2021)
	SM_1	I'm confident I can understand the most difficult learning material during my self-regulated learning with the help of generative AI.	(Daher & Abdela, 2025)
	SM_2	Even when study materials are uninteresting, I keep working until I finish with the help of generative AI.	(Pintrich & De Groot, 1990)
	SM_3	I persuade myself to work hard in SRL to improve my knowledge with the help of generative AI.	(Shen & Wang, 2025)
	SM_4	I try to connect the learning task to my personal interest with the help of generative AI.	(Shen & Wang, 2025)
Performance	SC_1	I make sure I keep up with the weekly plan of my self-regulated learning with the help of generative AI.	(Jansen et al., 2017)
	SC_2	Generative AI helps me stay focused on my learning tasks.	(Broadbent et al., 2023)

	SC_3	I use suitable learning strategies to improve my understanding during learning with the help of generative AI.	(Jansen et al., 2017)
	SC_4	I use generative AI to organise and manage my study time effectively.	(Jansen et al., 2017)
	SO_1	I use generative AI to check whether I am making progress in my learning.	(Panadero et al., 2021)
	SO_2	I use generative AI to evaluate how well I understand the learning materials.	(Jansen et al., 2017)
	SO_3	I monitor my learning progress when using generative AI and adjust my study strategies when necessary.	(Jansen et al., 2017)
	SO_4	While I perform a task, I check if the steps I am taking are appropriate with the help of generative AI.	(Panadero et al., 2021)
Self-Reflection Phase	SJ_1	I think about what I have learned after I finish my SRL task with the help of generative AI.	(Jansen et al., 2017)
	SJ_2	After completing a learning task, I use generative AI to evaluate whether I achieved my learning goals.	(Jansen et al., 2017)
	SJ_3	After learning, I use generative AI to compare my learning outcomes with my intended goals.	(Jansen et al., 2017)
	SJ_4	I find myself analysing the usefulness of my learning method while I used with the help of generative AI.	(Jansen et al., 2017)

SR_1	I adjust my learning strategies based on feedback with the help of generative AI.	(Pintrich et al., 1991; Broadbent et al., 2023)
SR_2	I use past mistakes to improve future learning with the help of generative AI.	(Pintrich et al., 1991; Broadbent et al., 2023)
SR_3	After the learning task completed, I asked myself on the way to improve in the next learning with the help of generative AI.	(Ebbes et al., 2024)
SR_4	I become motivated to learn other topics when my academic performance is improved with the help of generative AI.	(Shen & Wang, 2025)

Table 9

Section C: Perceived Academic Performance (Adjusted)

Item No.	Question	Reference
AP_1	I am confident that I can understand what I study and gain better performance in the exam.	(Ferla et al., 2010)
AP_2	I am able to complete my academic tasks effectively after my studies.	(Ferla et al., 2010)
AP_3	I believe I can achieve good academic performance in my studies.	(Boubker, 2024)

AP_4	I am satisfied with my overall academic performance after my studies.	(Boubker, 2024)

4.0 Results

4.1 Demographic Results

Table 10

Demographic Characteristics of the Respondents (n=222)

Characteristics	n	%
Gender		
n	222	100
Male	97	43.7
Female	125	56.3
Age		
16 - 18	20	9
19 - 25	184	82.9
26 - 30	12	5.4
31 and above	6	2.7
Educational Level		
Secondary School	24	10.8
Pre-U	14	6.3
Diploma	25	11.3
Undergraduate	153	68.9
Postgraduate	6	2.7

Table 9 presents the demographic characteristics of 222 respondents involved in this study. The sample had consisted of 97 male respondents (43.7%) and 125 female respondents (56.3%). This distribution shows that both genders were adequately represented in the study. For age,

most respondents (82.9%) were between 19 and 25 years old, followed by respondents aged 16–18 years (9.0%), 26–30 years (5.4%) and 31 years and above (2.7%). The sample is a good representation of the intended target population of students who are actively using generative AI for their learning. For educational level, undergraduate students formed the majority of the respondents (68.9%), followed by Diploma students (11.3%), Secondary School students (10.8%), pre-university students (6.3%) and Postgraduate students (2.7%). Overall, The respondent profile shows that the sample gathered is suitable for investigating how generative AI affects SRL and perceived academic achievement.

4.2 Descriptive Statistics

Table 11

Descriptive Statistics

Construct	Name	Mean	Standard deviation	Excess kurtosis	Skewness
Planning Affordance	PA_1	3.892	1.122	0.547	-1.076
	PA_2	4.005	1.016	-0.112	-0.814
	PA_3	4.086	0.952	0.716	-1.054
	PA_4	4.059	1.005	0.825	-1.137
	PA_5	3.820	1.164	-0.436	-0.731
Feedback & Interaction Affordance	FIA_1	4.405	0.734	2.078	-1.288
	FIA_2	4.266	0.820	2.033	-1.219
	FIA_3	4.270	0.753	0.845	-0.876
	FIA_4	4.257	0.891	2.382	-1.413
	FIA_5	4.117	1.011	0.922	-1.159
Forethought Phase	TA_1	3.950	0.978	0.488	-0.888
	TA_2	3.874	1.104	0.077	-0.881

	TA_3	3.959	1.019	0.493	-0.972
	TA_4	4.000	0.935	0.736	-0.933
	SM_1	4.162	0.748	1.285	-0.860
	SM_2	4.140	0.872	0.896	-0.933
	SM_3	4.027	0.925	1.456	-1.086
	SM_4	3.896	1.010	0.165	-0.819
Performance Phase	SC_1	4.095	1.007	1.117	-1.178
	SC_2	3.838	1.087	0.225	-0.882
	SC_3	4.086	0.873	1.639	-1.107
	SC_4	3.878	1.102	0.445	-0.976
	SO_1	4.081	1.006	1.242	-1.180
	SO_2	4.072	1.002	0.912	-1.120
	SO_3	4.014	0.966	0.718	-0.964
	SO_4	4.140	0.922	2.079	-1.324
Self-Reflection Phase	SJ_1	3.982	1.035	0.537	-0.994
	SJ_2	3.914	1.073	0.576	-1.019
	SJ_3	3.932	1.078	0.396	-0.973
	SJ_4	4.072	1.029	1.375	-1.271
	SR_1	4.054	1.051	1.156	-1.209
	SR_2	4.032	0.979	1.239	-1.136
	SR_3	3.878	1.130	0.255	-0.964
	SR_4	3.941	1.120	0.494	-1.047
Perceived Academic Performance	AP_1	4.158	0.831	0.944	-0.919
	AP_2	4.180	0.744	1.427	-0.900
	AP_3	4.176	0.754	0.196	-0.685

	AP_4	4.167	0.774	-0.249	-0.592

Table 11 presents the descriptive statistics for each measurement item, which include the mean, standard deviation, skewness and excess kurtosis. The mean scores from all measurement item are between 3.820 to 4.405. This indicate that respondents generally agreed with the statements that measuring generative AI affordances, SRL behaviours and perceived academic performance. This suggests that students generally perceived generative AI as beneficial in supporting their learning activities.

Among the Planning Affordance items, the mean values are between 3.820 to 4.086, where PA_3 recording the highest mean (4.086). This indicate that respondents generally agreed that generative AI helps them clarify learning requirements before beginning a task. For Feedback and Interaction Affordance, the mean values are between 4.117 to 4.405, with FIA_1 have the highest mean score (4.405). This suggest that students strongly agreed that AI explanations help them understand learning materials more effectively.

For the SRL constructs, the Forethought Phase recorded mean values between 3.874 and 4.162. Besides that, Performance Phase ranged from 3.838 to 4.140, Last but not least, the Self-Reflection Phase ranged from 3.878 to 4.072. These findings indicate that respondents generally demonstrated positive SRL behaviours when using generative AI throughout the planning, performance and reflection stages of learning.

For Perceived Academic Performance, all indicators achieved a high mean values between 4.158 to 4.180. This suggest that respondents generally perceived generative AI as contributing positively to their academic performance.

Overall, the descriptive statistics indicate that respondents had a positive perception

towards Generative AI affordances, SRL behaviours and perceived academic performance. Besides that, the standard deviation values are between 0.734 to 1.164, which indicate a moderate level of variation in respondent's answers. Furthermore, all skewness values were negative, where the responses were concentrated towards the "Agree" and "Strongly Agree" categories. Although several indicators recorded kurtosis values greater than 1, PLS-SEM is considered robust to moderate violations of the normality assumption (Hair et al., 2021). Therefore, the dataset was considered appropriate for subsequent PLS-SEM analysis.

4.3 Measurement Model

Table 12

Measurement Model

Construct	Item	Convergent Validity		Internal Consistency Reliability	
		Outer loadings	Average Variance Extracted (AVE)	Cronbach's Alpha	Composite Reliability
Planning Affordance	PA_1 ← PA	0.815	0.633	0.856	0.896
	PA_2 ← PA	0.770			
	PA_3 ← PA	0.765			
	PA_4 ← PA	0.794			
	PA_5 ← PA	0.831			
Feedback & Interaction Affordance	FIA_1 ← FIA	0.707	0.603	0.837	0.883
	FIA_2 ← FIA	0.808			
	FIA_3 ← FIA	0.747			
	FIA_4 ← FIA	0.828			
	FIA_5 ← FIA	0.786			

Forethought Phase	**TA_1 ← TA**	0.772	0.666	0.833	0.889
	TA_2 ← TA	0.844			
	TA_3 ← TA	0.821			
	TA_4 ← TA	0.827			
	SM_1 ← SM	0.748	0.611	0.787	0.862
	SM_2 ← SM	0.799			
	SM_3 ← SM	0.799			
	SM_4 ← SM	0.779			
Performance Phase	**SC_1 ← SC**	0.849	0.672	0.836	0.891
	SC_2 ← SC	0.857			
	SC_3 ← SC	0.835			
	SC_4 ← SC	0.731			
	SO_1 ← SO	0.786	0.658	0.827	0.885
	SO_2 ← SO	0.801			
	SO_3 ← SO	0.854			
	SO_4 ← SO	0.802			
Self-Reflection Phase	**SJ_1 ← SJ**	0.828	0.672	0.838	0.891
	SJ_2 ← SJ	0.846			
	SJ_3 ← SJ	0.800			
	SJ_4 ← SJ	0.805			
	SR_1 ← SR	0.750	0.696	0.854	0.901
	SR_2 ← SR	0.859			
	SR_3 ← SR	0.876			
	SR_4 ← SR	0.847			
Perceived	**AP_1 ← AP**	0.768	0.623	0.798	0.868

Academic Performance	**AP_2 ← AP**	0.734			
	AP_3 ← AP	0.825			
	AP_4 ← AP	0.827			

Table 13

Discriminant Validity (HTMT Value) Results

	AP	**FIA**	**PA**	**SC**	**SJ**	**SM**	**SO**	**SR**	**TA**
AP									
FIA	0.441								
PA	0.569	0.634							
SC	0.424	0.639	0.497						
SJ	0.396	0.619	0.678	0.768					
SM	0.680	0.652	0.680	0.617	0.633				
SO	0.530	0.712	0.590	0.807	0.745	0.709			
SR	0.529	0.616	0.698	0.759	0.839	0.751	0.755		
TA	0.368	0.656	0.555	0.728	0.720	0.801	0.676	0.717	

The measurement model was evaluated based on indicator reliability, internal consistency reliability, convergent validity and discriminant validity. The results are presented in **Tables 12 and 13.**

4.3.1 Indicator Reliability

Table 12 had presented the indicator reliability results based on the outer loadings values obtained from each measurement item. According to Hair et al. (2021), outer loading values of 0.7 or above indicate satisfactory indicator reliability. The results show that all retained indicators achieved outer loading values above 0.7, where the outer loadings are between 0.707 to 0.857 for affordance constructs, 0.748 to 0.844 for the Forethought Phase constructs, 0.731

to 0.857 for the Performance Phase constructs, 0.750 to 0.876 for the Self-Reflection Phase constructs and 0.734 to 0.827 for the Perceived Academic Performance construct. Thus, these findings indicate that all indicators demonstrated satisfactory indicator reliability and were retained for subsequent analyses.

4.3.2 Internal Consistency Reliability

Internal consistency reliability was evaluated using Cronbach's Alpha and Composite Reliability. According to Hair et al. (2021), Cronbach's Alpha values above 0.70 indicate satisfactory internal consistency. **Table 12** shows Cronbach's Alpha values range from 0.787 to 0.856 and Composite Reliability values range from 0.862 to 0.901 for all constructs. This means all values had exceeded the recommended threshold. Therefore, the constructs demonstrated satisfactory internal consistency reliability. The measurement items consistently measured their respective latent constructs.

4.3.3 Convergent Validity

Convergent validity was assessed using the Average Variance Extracted (AVE). from **Table 12**, all constructs achieved AVE values between 0.603 to 0.696, which exceed the recommended threshold of 0.50 (Hair et al., 2021). This indicates that each construct explains more than 50% of the variance of its measurement indicators. Therefore, the results confirm satisfactory convergent validity, indicating that the indicators within each construct adequately converge in measuring the same underlying concept.

4.3.4 Discriminant Validity

Discriminant validity was assessed using the HTMT. From **Table 13**, all HTMT values were below the recommended threshold (0.85), which were between 0.368 to 0.839. Therefore, discriminant validity was established, indicating that all constructs were empirically distinct

and measured different theoretical concepts. Therefore, adequate discriminant validity was established, which suggest that each construct possesses sufficient uniqueness and does not overlap excessively with other constructs in the model.

4.4 Structural Model

Table 14

Structural Model Result

Hypothesis Path		VIF	Effect Size	Hypothesis Testing			
			f^2	β	t-value	p-value	Decision
H1	PA → TA	1	0.302	0.482	7.7	<0.001	Supported
H2	PA → SM	1	0.461	0.562	9.286	<0.001	Supported
H3	FIA → SC	1	0.425	0.546	8.686	<0.001	Supported
H4	FIA → SO	1	0.601	0.613	10.735	<0.001	Supported
H5	FIA → SJ	1	0.414	0.541	10.707	<0.001	Supported
H6	FIA → SR	1	0.414	0.541	8.906	<0.001	Supported
H7	TA → AP	2.27	0.025	-0.194	2.219	0.027	**Not Supported**
H8	SM → AP	2.084	0.167	0.478	5.22	<0.001	Supported
H9	SC → AP	2.368	0.001	0.046	0.473	0.636	**Not Supported**
H10	SO → AP	2.282	0.019	0.171	2.1	0.036	Supported
H11	SJ → AP	2.447	0.002	-0.057	0.671	0.502	**Not Supported**
H12	SR → AP	2.659	0.015	0.163	1.337	0.181	**Not Supported**

Table 15

R-square and adjusted R-square

Endogenous Construct	R^2	Adjusted R^2
Academic Performance	0.344	0.325
Self-Control	0.298	0.295
Self-Judgement	0.293	0.29
Self-Motivation	0.316	0.313
Self-Observation	0.375	0.373
Self-Reaction	0.293	0.29
Task Analysis	0.232	0.229

Table 16

Q-square predict, RMSE and MAE

Indicator	Q^2predict	PLS-SEM		LM	
		RMSE	MAE	RMSE	MAE
AP_1	0.114	0.787	0.618	0.819	0.638
AP_2	0.089	0.714	0.539	0.733	0.566
AP_3	0.126	0.708	0.562	0.734	0.597
AP_4	0.139	0.722	0.579	0.735	0.587
SC_1	0.198	0.905	0.672	0.956	0.7
SC_2	0.188	0.984	0.766	1.035	0.796
SC_3	0.24	0.765	0.588	0.781	0.599
SC_4	0.136	1.03	0.8	1.016	0.766
SJ_1	0.246	0.903	0.699	0.903	0.692

SJ_2	0.141	0.998	0.762	0.976	0.729
SJ_3	0.154	0.995	0.761	0.916	0.682
SJ_4	0.19	0.931	0.702	0.92	0.684
SM_1	0.172	0.684	0.541	0.703	0.55
SM_2	0.167	0.799	0.625	0.827	0.638
SM_3	0.2	0.831	0.628	0.807	0.599
SM_4	0.194	0.912	0.706	0.898	0.703
SO_1	0.199	0.904	0.687	0.887	0.664
SO_2	0.212	0.894	0.68	0.904	0.673
SO_3	0.221	0.856	0.657	0.827	0.633
SO_4	0.296	0.777	0.571	0.798	0.59
SR_1	0.127	0.987	0.741	0.939	0.7
SR_2	0.227	0.864	0.649	0.868	0.621
SR_3	0.231	0.995	0.75	0.928	0.671
SR_4	0.199	1.005	0.747	0.984	0.723
TA_1	0.108	0.929	0.729	0.854	0.655
TA_2	0.14	1.029	0.809	0.965	0.749
TA_3	0.127	0.956	0.733	0.884	0.657
TA_4	0.197	0.842	0.651	0.81	0.629

The structural model was evaluated by examining collinearity issues, hypothesis testing, explanatory power, and predictive power. The results are presented **in Tables 14, 15, and 16.**

4.4.1 Collinearity Issues

Collinearity was assessed using the VIF. From **Table 14**, all VIF values are between 1 to

2.659, which are well below the recommended threshold of 5.0 (Hair et al., 2021). Therefore, there were no multicollinearity issues identified among the predictor constructs. The predictor constructs do not exhibit excessive correlations that could bias the estimation of the structural relationships. Therefore, the structural model was considered suitable for hypothesis testing.

4.4.2 Hypothesis Testing

The significance of the structural relationships was evaluated using path coefficients (β), t-values and p-values obtained through the bootstrapping in SmartPLS 4. The results showed that eight hypotheses were supported. Specifically, Planning Affordance had a strong positive effect on Task Analysis (β = 0.482, $p < 0.001$) and Self-Motivation (β = 0.562, $p < 0.001$). This indicates that when students find Generative AI as useful tool for planning, they are more likely better at analyse learning tasks and stay learning motivation before academic activities start. Similarly, Feedback and Interaction Affordance have a positive effect Self-Control (β = 0.546, $p < 0.001$), Self-Observation (β = 0.613, $p < 0.001$), Self-Judgement (β = 0.541, $p < 0.001$) and Self-Reaction (β = 0.541, $p < 0.001$). This suggests that timely feedback and interactive support provided by Generative AI contribute positively to student's SRL behaviours throughout the learning process.

Among the SRL dimensions predicting Academic Performance, Self-Motivation (β = 0.478, p < 0.001) and Self-Observation (β = 0.171, p = 0.036) demonstrated significant positive effects that supporting the proposed hypotheses. This indicate that student who keep monitor their learning progress will be more likely to achieve a better academic performance compared to other. However, Task Analysis showed a significant negative effect on Academic Performance (β = –0.194, p = 0.027), where having a relationship in the opposite direction to that hypothesized. Therefore, it did not support the proposed hypothesis. Finally, Self-Control (β =

0.046, p = 0.636), Self-Judgement (β = –0.057, p = 0.502) and Self-Reaction (β = 0.163, p = 0.181) did not showed a significant relationship with Academic Performance (p > 0.05). Therefore, the corresponding hypotheses for these constructs were rejected.

4.4.3 Explanatory Power

The explanatory power of the structural model was assessed using the coefficient of determination (R^2). From **Table 15**, Academic Performance achieved an R^2 value of 0.344, which mean there are 34.4% of the variance in Academic Performance was explained by the SRL constructs. Besides that, Self-Observation recorded the highest explanatory power (R^2 = 0.375), followed by Self-Motivation (R^2 = 0.316), Self-Control (R^2 = 0.298), Self-Judgement (R^2 = 0.293), Self-Reaction (R^2 = 0.293) and Task Analysis (R^2 = 0.232). According to Hair et al. (2021), these R^2 values indicate a moderate (between 0.25 and 0.49) and weak (below 0.25) level of explanatory power. Therefore, the proposed structural model adequately explains the endogenous constructs included in this study.

4.4.4 Effect Size

The effect size (f^2) values are presented in Table 14. According to Cohen's (1988) guidelines, Planning Affordance exhibited medium-to-large effects on Task Analysis (f^2 = 0.302) and Self-Motivation (f^2 = 0.461). Similarly, Feedback and Interaction Affordance demonstrated medium-to-large effects on Self-Control (f^2 = 0.425), Self-Observation (f^2 = 0.601), Self-Judgement (f^2 = 0.414) and Self-Reaction (f^2 = 0.414). H However, the influences of the SRL dimensions on Academic Performance ranged from non-significant to small, with Self-Motivation having the highest contribution (f^2 = 0.167). The results suggest that the affordances of Generative AI have a significant impact on students' SRL behaviours, while the direct contribution of the dimensions of SRL to Academic Performance is relatively low.

4.5 Predictive Power

Predictive power was evaluated using PLSpredict. From **Table 16**, all Q^2_predict values were positive and located between 0.089 to 0.296. This indicate that the proposed model possesses predictive relevance. Furthermore, the PLS-SEM model produced lower RMSE and MAE values than the LM for most indicators. This indicates superior out-of-sample predictive performance. Overall, the findings confirm that the proposed PLS-SEM model possesses robust predictive power. It successfully explaining and predicting student's SRL behaviours and perceived academic performance.

5.0 Discussion

5.1 Introduction

This chapter discusses the findings obtained from the PLS-SEM analysis by relating them to the research objectives, Affordance Theory, Zimmerman's SRL theory and previous empirical studies. This chapter will interpret the findings to explain how Generative AI affordances influence student's SRL behaviours and perceived academic performance. The discussion is organised according to the three research objectives, followed by a summary of the key findings and their implications.

5.2 Discussion of Research Objective 1

The first objective of this research is to examine the relationship between Generative AI affordances and student's SRL. The finding demonstrated that Planning Affordance and Feedback and Interaction Affordance positively influenced the dimensions of SRL. Specifically, Planning Affordance positively influenced Task Analysis and Self-Motivation, while Feedback and Interaction Affordance significantly influenced Self-Control, Self-Observation, Self-Judgement and Self-Reaction. Therefore, the first objective was achieved.

5.2.1 Planning Affordance and Task Analysis

The results showed that there is a positive relationship between Planning Affordance and Task Analysis. This finding suggests that students perceive Generative AI as an effective learning assistant that helps them clarify learning objectives, understand task requirements and organise learning activities before beginning academic tasks. Rather than immediately searching for answers, students will first use Generative AI to structure their learning process, understand the task expectations and make a appropriate learning strategies for them. Consequently, Generative AI assists students in preparing for learning more systematically and efficiently.

This finding is consistent with Daher and Abdela (2025) who reported that Generative AI facilitates learning planning by helping the student to organise their learning tasks, explain academic requirements and provide recommended learning strategies to them. Similarly, Zimmerman (2002) proposed that effective SRL begins with the forethought phase, where the learners will analyse the task requirements and establish learning goals before starting in learning activities. The present findings therefore suggest that Generative AI effectively supports this planning stage by providing students with structured guidance before they start on their studies.

Furthermore, these findings demonstrate the importance of Planning Affordance within affordance theory. Based on the theory, technological features only become meaningful when users actively interact with them to accomplish specific goal. In this study, student transformed the planning functions provided by Generative AI into meaningful learning support by using AI to organise their learning activities before they start the academic tasks.

5.2.2 Planning Affordance and Self-Motivation

The findings also demonstrated that Planning Affordance positively influenced Self-Motivation. Students who use Generative AI to organise their learning activities reported higher motivation toward completing their academic tasks. The personalised study recommendation and structured planning help to reduce uncertainty before they start to learning. This allows students to approach their learning activities with higher confidence. When students clearly understand what they need to accomplish and receive appropriate guidance, they will become more willing to engage with challenging learning materials and sustain their efforts.

This finding supports Shu and Xu (2022), who found that AI-supported learning will enhances student's autonomous learning and learning motivation. They reported that student who interact more frequently with AI tend to develop a stronger SRL strategies. Similarly, Zimmerman (2002) argued that motivational beliefs such as self-efficacy and intrinsic interest are fundamental components of the Forethought Phase because they determine the learner's willingness to initiate and sustain learning activities. Therefore, the present study suggests that planning capabilities of Generative AI help to strengthen student's motivational beliefs by enabling them to become more actively involved in the learning process. Moreover, these finding also indicate that the planning capabilities offered by Generative AI create an opportunity for students who actively utilise AI-generated suggestion to organise and prepare their learning activities.

5.2.3 Feedback and Interaction Affordance and the Performance Phase

The finding further revealed that Feedback and Interaction Affordance significantly influenced Self-Control and Self-Observation. This indicates that students actively use Generative Ai not only obtain the answers but also to monitor their understanding, evaluate their progress and regulate their learning behaviour throughout the learning process. Unlike

conventional learning resources that provide static information, Generative AI will provides immediate and personalised feedback that allow students to identify misunderstandings and adjust their learning strategies.

Among these relationships, the strongest effect was between Feedback and Interaction Affordance and Self-Observation. This finding suggests that the continuous interaction provided by Generative AI encourages students to monitor their learning progress more frequently. The immediate explanations, personalised feedback and interactive conversation from Generative AI help learner to recognise their learning weaknesses early and make appropriate adjustments before misconceptions become more serious. As a result, students become more aware of their own learning progress and able to regulate their learning behaviours better.

These findings are aligned with Ng et al. (2024), where they reported that AI-generated feedback promotes reflective thinking, behavioural engagement and continuous self-monitoring to learners. Similarly, Zimmerman (2002) emphasised that Self-Observation enables learners to monitor their own learning behaviours and evaluate whether their learning strategies are effective throughout their learning process. Therefore, the finding shows that Feedback and Interaction Affordance effectively in strengthens student's ability to regulate their learning during the Performance Phase.

Similarly, the positive relationship between Feedback and Interaction Affordance and Self-Control shows that personalised AI feedback helps students maintain concentration, regulate their study behaviours and apply appropriate learning strategies while doing an academic task. Instead of depending on external guidance, students are able to make continuous adjustments based on the immediate feedback from generative AI, which lead to improve

learning efficiency.

5.2.4 Feedback and Interaction Affordance and the Self-Reflection Phase

The findings also demonstrated that feedback and Interaction Affordance positively influenced Self-Judgement and Self-Reaction. These findings suggest that students use the feedback generated by generative AI to evaluate whether they have achieved their learning goal to reflect on the effectiveness of their learning strategies after complete an academic task.

The immediate AI-generated feedback provides student an opportunity to compare their performance against expected learning outcomes and find areas that need further improvement. Consequently, students will become more capable of evaluating their own learning performance and making informed decisions for their next learning strategy.

These finding support Zimmerman (2002) who described self-judgement as learner's evaluation of their own learning performance through self-assessment and casual attribution. Likewise, self-reaction represents learner's emotional and behavioural responses after evaluating their learning performance. This enables them to adapt their future learning strategies. Besides that, the findings also consistent with Ge et al. (2025) who found that AI-supported reflection systems improve learner's metacognitive awareness by encouraging continuous evaluation and adaptation of learning strategies. Therefore, Generative AI appears to support reflective learning by providing timely feedback that facilities continuous improvement throughout the learning process.

5.3 Discussion of Research Objective 2

The second objective of this research is to examine the relationship between SRL and

perceived academic performance. The finding demonstrated that only Self-Motivation and Self-Observation significantly influenced academic performance. Task analysis, self-control, self-judgement and self-reaction did not support the proposed hypotheses. These finding suggest that although Generative AI enhance various dimensions of SRL, but not every SRL behaviour will provide positive effect on improving academic performance. Therefore, the second research objective was only partially achieved.

5.3.1 Self-Motivation and Perceived Academic Performance

Among all dimensions of self-regulated learning, Self-Motivation emerged as the strongest predictor of perceived academic performance. This finding suggests that students with stronger learning motivation are likely to actively involve in academic activities and utilise Generative AI as a learning companion rather than merely an answer-generation tool. The motivated learner generally more willing to seek for clarification, explore more learning resources and persist when facing challenging academic tasks. Consequently, they are more likely to perceive improvements in their academic performance.

The finding is consistent with Zimmerman's (2002) SRL theory where the motivational beliefs such as self-efficacy, intrinsic interest and outcome expectation are fundamental element of Forethought Phase that influence learner's willingness to initiate and sustain learning activities. Therefore, the findings of the present study suggest that Generative AI indirectly contributes to improved academic performance by strengthening students' motivation to engage actively in learning.

5.3.2 Self-Observation and Perceived Academic Performance

Self-Observation was also found to have a significant positive influence on perceived

academic performance. This finding indicates that students who always monitor their learning progress are better able to identify the misunderstandings, evaluate their learning performance and modify their learning strategies when needed. Rather than waiting until an academic task done, the student will actively evaluate their understanding during the learning process with the help of Generative AI.

The immediate and personalised feedback enables students to detect their learning weaknesses early and make appropriate adjustments before misconceptions become more serious. As a results, students become more aware of their own learning progress, which contributes to better learning outcomes and higher perceived academic performance.

These findings are consistent with Ng et al. (2024) who reported that AI-generated feedback promotes reflective thinking and continuous self-monitoring among learners. Similarly, Zimmerman (2002) explained that self-observation enables learners to monitor their learning behaviours continuously and evaluate whether their learning strategies a remain effective throughout the learning process. Therefore, continuous monitoring supported by generative AI play an important role in improving student's academic performance.

5.3.3 Task Analysis and Perceived Academic Performance

Although Task Analysis demonstrates a significant relationship with perceived academic performance, but it has the opposite direction to the proposed hypothesis, which provide negative effect. Although Task Analysis is theoretically considered an essential component of the Forethought Phase, the finding suggests that analysing learning tasks alone does not improve student's academic performance.

The possible reason is because students who spend time in planning and analysing

learning tasks may delay the actual learning process. While careful planning helps students understand learning objectives, but excessive planning without effective implementation may reduce learning efficiency. Consequently, students may invest substantial effort in preparing for learning but fail to translate those plans into effective learning behaviours.

Another possible explanation is that students facing more difficult or complex academic tasks naturally perform more extensive task analysis than students completing relatively simple learning activities. As a result, this may lead these students to feel that their academic performance is lower, even when they plan more in detail. The results suggest that planning alone is not sufficient to improve academic performance but must be complemented by effective implementation and continuous monitoring throughout the learning process.

Although this finding differs from the proposed hypothesis, it highlights that successful learning depends not only on careful planning but also on student's ability to implement appropriate learning strategies throughout the learning process.

5.3.4 Self-Control, Self-Judgement and Self-Reaction

The findings further revealed that Self-Control, Self-Judgement and Self-Reaction did not significantly influence perceived academic performance. Although these dimensions are recognised as important components of Zimmerman's SRL theory, but they did not demonstrate direct relationships with student's perceived academic performance in the present study.

One of the possible explanations is that maintaining concentration and regulating learning behaviours alone may not necessarily improve academic performance. Students may successfully minimise distractions and maintain focus while studying, but these behaviours

may not translate into better academic outcomes unless they are accompanied by effective cognitive processing and meaningful learning activities.

Similarly, Self-Judgement and Self-Reaction represent reflective processes that occur after learning activities have been completed. Students may evaluate their learning performance, recognise their weaknesses and identify areas requiring improvement. However, these reflections may not immediately contribute to improved academic performance unless students subsequently apply the lessons learned to future learning experiences.

5.4 Discussion of Research Objective 3

The third objective of this study was to validate the proposed conceptual model by examining the relationships among generative AI affordances, SRL and academic performance. Overall, the findings found that the proposed model is reliable and demonstrates satisfactory predictive capability and successfully achieve this research objective.

The descriptive analysis had provided an overview of student's perceptions of generative AI, SRL and academic performance. The mean scores for all constructs were located between 3.820 and 4.405. This indicate that respondents generally agreed with the survey statements. This suggests that students have positive perceptions of the planning and feedback affordances provided by generative AI and actively practice SRL behaviours during their studies. In addition, the relatively high mean scores for academic performance indicate that students generally perceived generative AI are useful in supporting their learning.

The structural model further examined the relationships among the study constructs. The significant path coefficients indicate positive relationships between generative AI affordances

and SRL, which suggest that students who perceived greater planning and feedback support from generative AI were more likely to demonstrate a stronger self-regulated learning behaviour. Specifically, Feedback and Interaction Affordance showed a strong positive relationship with Self-Observation ($\beta = 0.613$), highlighting the importance of AI-generated feedback in encouraging students to monitor and reflect on their learning. Furthermore, the results showed that Self-Motivation ($\beta = 0.478$) and Self-Observation ($\beta = 0.171$) had significant positive relationships with academic performance, indicating that these SRL components play an important role to improve their academic outcomes.

Following the examination of these relationships, the measurement and structural models were evaluated to validate the proposed conceptual model. The measurement model had achieved satisfactory level for all indicator reliability, internal consistency reliability, convergent validity and discriminant validity. This help to confirm that the constructs were measured accurately. Besides that, the structural model also demonstrated satisfactory explanatory and predictive capability. The model explained 34.4% of the variance in perceived academic performance, while the positive Q^2predict values and lower prediction errors confirmed that the model possesses acceptable out-of-sample predictive ability.

Although most constructs had observed positive relationships, there were two findings differed from the proposed hypotheses. Task Analysis demonstrated a significant negative relationship with academic performance, which suggested that planning and analysing learning tasks does not always lead to better academic outcomes. One possible reason is students who spend excessive time planning may delay the implementation of effective learning strategies or their plan increased their cognitive load that led to low learning efficiency. In addition, Self-Judgement showed a negative but non-significant relationship with academic performance. Although the coefficient was negative, the relationship was not statistically significant. This

indicate that there is insufficient evidence to conclude that Self-Judgement negatively influences academic performance. This finding suggests that evaluating own learning alone may not directly improve academic performance unless students use the evaluation results to adjust and improve their learning strategies.

Overall, the validated conceptual model demonstrates that generative AI affordances contribute to student's academic performance primarily by supporting the development of SRL behaviours. Rather than directly improving their academic outcomes, generative AI serves as a learning support tool that encourages students to plan, monitor, and reflect on their learning. This ultimately helps students achieve better academic performance. Therefore, the proposed conceptual model provides a useful framework for understanding how generative AI can be effectively integrated into higher education to support students' learning and academic success.

5.5 Recommendations

Based on the findings of this study, several recommendations are proposed for higher education institutions, educators, students, developers of generative AI applications, and future researchers to maximise the educational benefits of generative AI.

Firstly, higher education institutions should provide training workshops to promote the effective and responsible use of generative AI for learning. Universities should strengthen the way of student using generative AI to support their learning activities such as planning study tasks and monitoring learning progress, rather than asking AI to generate answers. By encouraging the correct way to use generative AI, these can enhance their SRL skills while promoting responsible AI usage and maintaining academic integrity.

Besides that, lecturer is encouraged to integrate generative AI into teaching and learning activities in ways to promote SRL. Since self-motivation and self-observation have significant positive relationships to academic performance, lecturer's can design learning activities that encourage students to use generative AI to monitor their learning progress, evaluate their understanding and identify areas for improvement. In addition, lecturer can also guide students to critically evaluate AI-generated responses instead of accepting them without verification. This method can encourage student to have independent learning, critical thinking and continuous reflection throughout the learning process.

Moreover, generative AI developer should continue improving AI features that support students' SRL. The findings had demonstrated that Feedback and Interaction Affordance have the strongest relationship with Self-Observation, which indicates the importance of instant and personalised feedback. Therefore, AI developers should enhance adaptive feedback, personalised learning recommendations, progress tracking and reflective prompts that encourage students to use for monitoring their learning. Thus, these improvements may help students develop stronger SRL behaviours and achieve better academic outcomes.

Finally, future researchers can extend this study by including students from different educational institutions, academic disciplines and countries to improve the generalisability of the findings. Longitudinal studies should also be conducted to examine how generative AI influences student's SRL and academic performance over time. In addition, future studies should also conduct mediation analysis to examine whether self-regulated learning mediates the relationship between generative AI affordances and academic performance. This will provide a clearer understanding of the indirect effects of generative AI on student's academic outcomes.

Overall, these recommendations will provide practical guidance for higher education institutions, lecturer, AI developers and researchers to promote the effective use of generative AI to strengthen student's SRL and enhance their academic performance.

6.0 Conclusion

This study examined the relationships among generative AI affordances, SRL and academic performance using Affordance Theory and Zimmerman's SRL framework. By using PLS-SEM, the proposed conceptual model was successfully validated by providing empirical evidence on how generative AI supports student's learning processes and academic outcomes. The findings had achieved all three research objectives and demonstrated how generative AI can effectively support different phases of SRL when used appropriately. The findings showed that Planning Affordance positively influenced Task Analysis and Self-Motivation, while Feedback and Interaction Affordance positively influenced Self-Control, Self-Observation, Self-Judgement and Self-Reaction. Furthermore, Self-Motivation and Self-Observation were found to have significant positive relationships with perceived academic performance. The validated measurement and structural models further confirmed that the proposed conceptual model has satisfactory reliability, validity, and predictive capability. Overall, this study contributes to the understanding of how generative AI effectively integrated into education to support student's SRL and improve academic performance. The findings also provide practical guidance for higher education institutions, educators, AI developers and future researchers in promoting the responsible use of generative AI to enhance learning outcomes. As generative AI continues to evolve, its effective integration into education has the potential to foster more independent, motivated, and reflective learners.

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